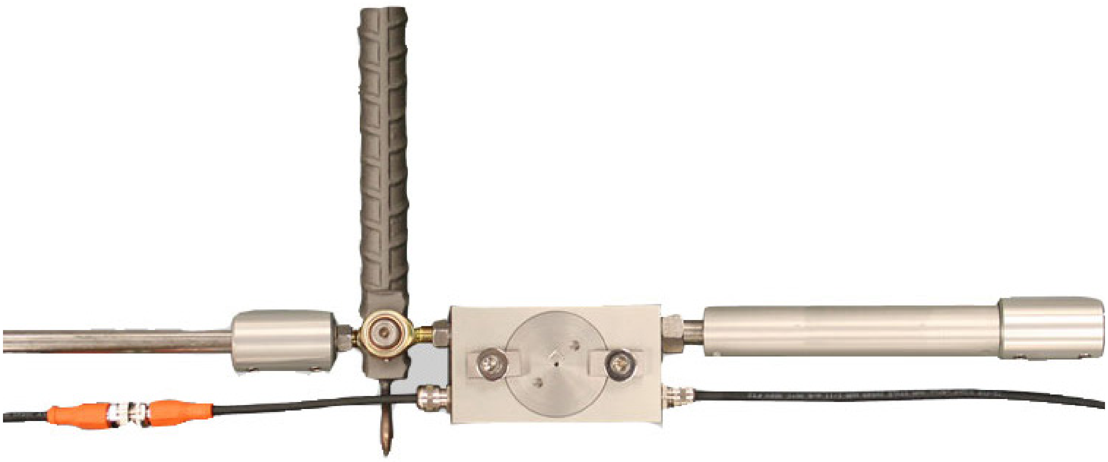
DATASHEET

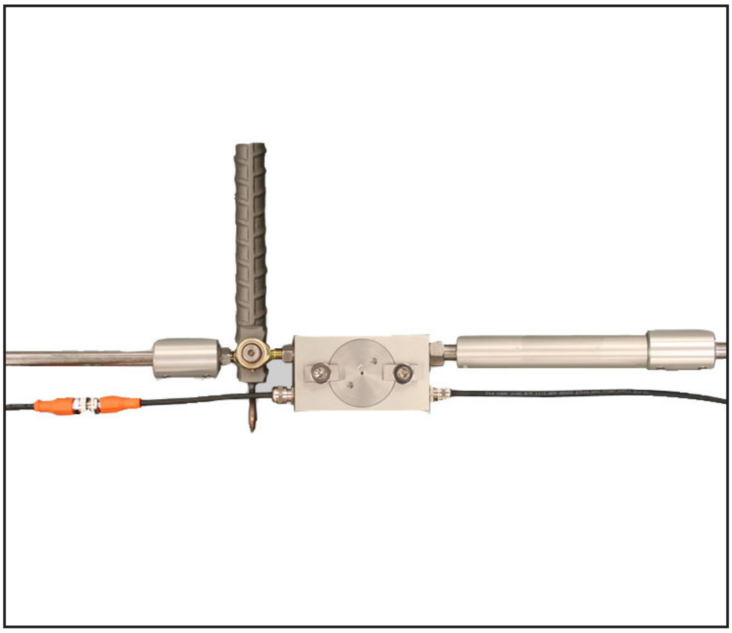
Tunnel Profile Monitoring System

Used to monitor strain along stranded anchors. The low profile design means that multiple sensors can be installed along the bonded length of strand anchors



Tunnel Profile Monitoring System

Overview



The Tunnel Profile Monitoring System is a series of linked rods fixed to the tunnel wall to monitor deformation. A data logging system and related software is available to provide near real time displacement and generate a graphical representation of the tunnel performance.

A system of linked arms is affixed to the tunnel wall. Each arm is fitted with a high accuracy displacement sensor and precision tilt meter. Spatial displacement of the pins and arms results in changed tilt and displacement readings. The data logger system automatically collects the data and transmits it to a computer. The computer then analyses the data and calculates the displacement profile for presentation.

The system is available in either open or closed loop configuration. The closed loop method is analogous to conventional closed end survey techniques, while the open loop must be referenced to a know location.

APPLICATIONS

- Monitor underground openings during construction for control and safety
- Monitor tunnel deformation due to nearby construction activities
- Monitor long-term deformation and performance of existing tunnels

FEATURES

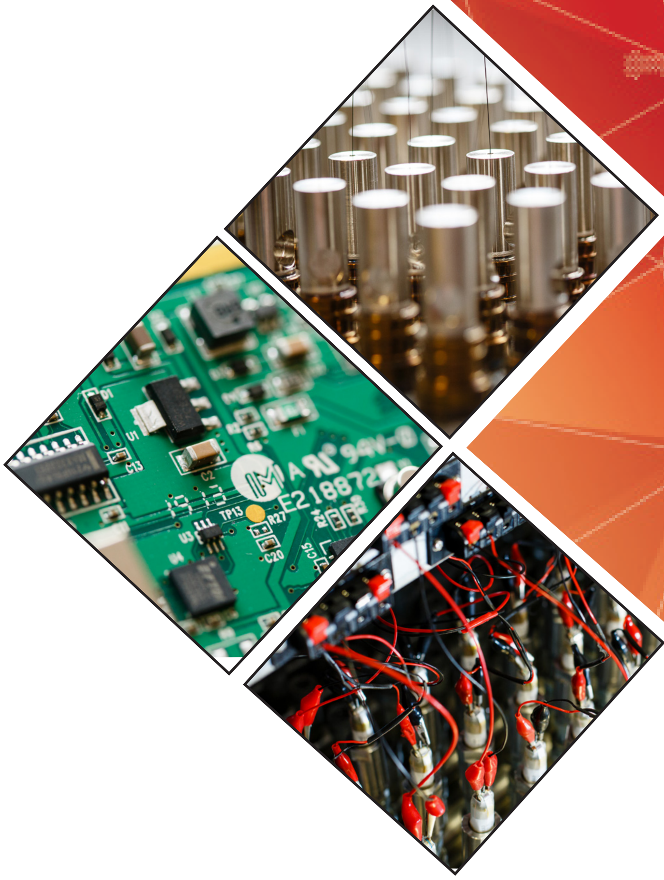
- Very low profile design, suitable for installation in the tight space available around TBMs
- No tunnel traffic interference
- Digital Bussed: single cable per arm to simplify installation and reduce cost
- Direct measurement of displacement rather than calculating displacement from a tilt measurement
- Immune to the air density related problems inherent in optical systems
- On-board electronics to minimise electrical noise problems, and permit tilt sensor calibration independent of cable length effects. Cable length may be changed without affecting sensor calibration
- Near-real time software with full graphic and alarm capability
- Built-in connectors for manual tape extensometer connection to verify operation, and aid in initial installation and commissioning

CE MEMS

Tunnel Profile Monitoring System

Specifications

DISPLACEMENT SENSOR	
Total Mechanical Travel	25mm
Shock	50 g 11 ms half sine
Vibration	20 g ms 5 Hz to 2 KHz
Life	One billion operations
Independent Linearity	0.25%
Operating Temperature	-40 to 80°C
Resolution	Infinite
Accuracy	0.06mm
TILT SENSOR SPECIFICATIONS	
Range	±15° (other ranges upon request)
Resolution	±2 arc sec. (±0.0006°) (0.01 mm/m)
Nonlinearity	±0.0125% FS (±0.002°) (0.03 mm/m)
Repeatability	±0.0125% FS (±0.002°) (0.03 mm/m)
Sensor	MEMS (Micro-Electro-Mechanical Systems) Accelerometer, Uniaxial
Operating Temperature	-40 to 85°C
Cable	Type 800 - Multi-core with Braid
SYSTEM COMPONENTS	
Tilt/displacement sensor assembly	
Extension tube	
Electrical cable sensor to logger	
Reference pin comes with tape extensometer connector	
Data logger system	
GeoAxiom software	
Manual	



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